

Visvesvaraya Technological University, Belagavi
Model Question Paper-2 with effect from 2022
Electrical and Electronics Engg. Stream (CBCS Scheme)
First /Second Semester Engineering Degree Examination

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Subject Title: Chemistry for Electrical & Electronics Engg. Stream 22CHEE12/22

TIME: 03 Hours

Max. Marks: 100

Note: Answer any **FIVE** full questions, choosing at least **ONE** question from each MODULE.

MODULE 1			Marks
Q.1	a	What are conductors, insulators and semiconductors explain with an example for each.	7
	b	Describe the purification of electronic grade silicon from quartz by Float Zone method	6
	c	Explain the synthesis of polyacetylene by Zeigler Natta catalyst, along with its conducting mechanism	7
OR			
Q.2	a	Define the following (i) conducting polymers (ii) Number average molecular weight (iii) Weight average molecular weight.	6
	b	What is electroless plating? Describe electroless plating of copper in the manufacture of double - sided PCB.	7
	c	A polydisperse sample of polystyrene is prepared by mixing three monodisperse samples in the following proportions. 10g of 1000 molecular weight, 20g of 40000 molecular weight and 30g of 50000 molecular weight. Determine number average and weight average molecular weight. Find the index of polydispersity.	7
MODULE 2			
Q.3	a	What are batteries? Explain the different components of a battery examples.	6
	b	What are fuel cells? Explain the construction and working of polymer-oxygen fuel cell.	7
	c	Explain the construction and working of Vanadium flow battery. Mention its applications.	7
OR			
Q.4	a	Explain the construction and working of Lithium-polymer battery. Mention its applications.	7
	b	What are photovoltaic cells? Describe the construction and working of a photovoltaic cell. Mention its advantages and disadvantages.	7
	c	What are fuel cells? What are batteries? Mention few differences between them.	6

MODULE 3			
Q.5	a	What is metallic corrosion? Explain its technological importance	7
	b	Explain Anodization of Aluminium, and mention its applications..	7
	c	What is CPR? A thick brass sheet of area 100 inch ² is exposed to moist air. After 1 years of period, it was found to experience a weight loss 100 g due to corrosion. If the density of brass is 8.73 g/cm ³ . Calculate CPR in mpy and mmpy.	6
OR			
Q.6	a	Explain differential metal corrosion with an example	6
	b	Describe galvanizing and mention its applications.	7
	c	What is e-waste? Describe the effects of e-waste on environment and human health	7
MODULE 4			
Q.7	a	What are nanomaterials? Explain effect of size on Surface area, conducting properties of nanomaterials.	7
	b	Write a note on nanophotonics and nanosensors.	6
	c	What are QLED'S? Mention its properties along with their applications.	7
OR			
Q.8	a	What are OLED'S? Mention its properties along with their applications.	7
	b	Describe the synthesis of nanomaterials by Co-precipitation method with example.	7
	c	What are perovskite materials? Give the properties and applications of perovskite materials in optoelectronic devices.	6
MODULE 5			
Q.9	a	What are reference electrodes? Explain the construction and working principle of Calomel electrode.	7
	b	Explain the working principle and applications of colorimetric sensor.	7
	c	What are concentration cells? Represent a cell formed by immersing two silver electrodes in silver nitrate solutions of concentration 0.05 and 0.2 M. Write the reactions and find the emf of the cell.	6
OR			
Q.10	a	What are Conductometric sensors? Explain the principle, instrumentation and the application of conductometric sensor in the estimation of weak acids.	7
	b	Mention different types of electrodes? Describe how Ph of the given electrode measured using glass electrode.	7
	c	Explain how the strength of a weak acid determined using a conductometric sensor.	6